

Where the rain goes

Speaker notes — Al-Masar Al-Akhdar green corridor, Beirut

These notes accompany the slide deck. Each entry gives the argument to make, the numbers behind it, and the answers to the questions that slide is most likely to attract. Read the *Key numbers* table at the end before presenting — everything quoted on the slides is drawn from it.

SLIDE 1

Where the rain goes

Opening. This is a physics-based flood model of the Fouad Boutros right-of-way in Beirut, built from a 39 GB airborne LiDAR survey and run at half-metre resolution. The talk has three parts: how the model works, how far it can be trusted, and what it says about the proposed green corridor.

SLIDE 2

Chapter: how the simulator works

Section break.

SLIDE 3

Rain falls on every cell

The key modelling choice is rain-on-grid: rainfall is applied to every cell and the water finds its own way downhill. Nothing is pre-routed, so where flooding appears is a result rather than an input.

Each cell carries its own infiltration rate, surface roughness and detention depth, derived from a nine-class land-cover classification of the LiDAR returns and aerial imagery. Buildings stand as solid obstacles at roof height, and the rain landing on roofs and enclosed courtyards is rerouted to the nearest street cell, which is what a downspout does.

The figure shows the three design storms. The 25 November 2025 event is real: 25.4 mm in 30 minutes. By the Lebanese stormwater code's own intensity table that is only a 2-5 year burst, yet it flooded Sassine Square and the Ring. The flooding problem here is not extreme rainfall; it is ordinary rain on steep, sealed streets.

Mass balance is checked every run — rain in equals infiltration plus drainage plus outflow plus storage, closing to about one part in 30,000. A model that loses water silently can produce any answer you like.

SLIDE 4

The corridor, surface by surface

The design is not a polygon with an average infiltration rate. Each of the seven surface types in the published cross-section — vehicular lane, porous bikelane, bioswale, porous sidewalk, rain garden, terrace, bioretention pond — is written into the terrain cell by cell with its own infiltration rate, Manning roughness and detention depth.

Worth knowing before someone asks which element does the work: rain garden is over half the corridor by area. The specialist detention features, the bioswales and bioretention ponds, together come to under 4,000 m². Most of the benefit comes from plain permeable ground, not from engineered structures.

The corridor also demolishes buildings in its path. Those are identified from the official highway alignment, excluding heritage and institutional buildings which the real scheme preserves and repurposes, and cleared from the terrain before the corridor is laid down.

IMPORTANT TO SAY OUT LOUD, and do not wait to be asked. The internal structure of the corridor here is our interpretation. We had the official zone polygon and the right-of-way alignment as data, but the cross-section — how wide the bikelane is, where the bioswale sits relative to the sidewalk, where the ponds and terraces go — was read off the published design drawings by eye and completed with what seemed hydraulically sensible. Band widths follow the drawing as we read it; bioretention ponds are placed at terrain low points and terraces on the steep segments, which is standard practice and also where our own model independently shows water collecting.

So this is a faithful-in-spirit representation, not a construction specification. If the design team can share the actual cross-section dimensions and element positions, we can drop them straight in and re-run — the pipeline takes a material raster, so it is a data swap rather than a rebuild. The headline conclusions are unlikely to move much, because they are driven by total permeable area and where the corridor sits in the catchment rather than by the exact arrangement within it — but that is an expectation, not a tested claim.

SLIDE 5

Chapter: validation

Section break.

SLIDE 6

Against an independent solver

This is the strongest evidence in the deck. SynxFlow is a separate piece of software, written by a different group, solving the full shallow-water equations with a different numerical scheme. We ran it on the identical 0.5 m corridor grid and the identical storm.

Our shock-capturing scheme agrees with it to an extent IoU of 0.913 and a depth RMSE of 3.7 cm. For context, the UK Environment Agency's own benchmark exercise saw worse agreement between commercial packages on a much gentler test case.

The comparison also exposed something about our own fast scheme: it spreads water over 27 % more area than the full solver. That is a known characteristic of simplified three-term schemes — they smear the flood front because they drop the advective momentum term. It matters here because these streets run supercritical at 6-8 m/s on a 6 % grade.

A third engine, LISFLOOD-FP, could not be used at all. On this stepped terrain it fabricated roughly six times the input water volume and failed its own mass balance check, so its output is void rather than merely different. That is a documented limitation of its scheme on terrain with retaining walls, not a bug we introduced.

SLIDE 7

Benchmark: Test 8A

Of the eight EA benchmark cases this is the one closest to what we actually do: rain applied directly onto a real, dense urban street network — the Glasgow test site — with no inflow hydrograph and no drainage network.

The grey bars show the published spread of the 19 packages that took part. The darker inner bar is what the report calls the main cluster: it excludes the simplified 'volume-spreading' codes that the report itself describes as approximate. Being inside the cluster is the stronger claim, and we are inside it at all four gauges where the report publishes curves.

Only four of the nine output points have published time series in the report, which is why the figure shows four. Our values at the other five are in the results file but there is nothing to compare them against.

One caveat to give if pressed: the published bands here were read off the report's figures rather than obtained as data, so they are good to a centimetre or two. That is not enough to change the verdict — our points sit well inside the bands, not on their edges.

SLIDE 8

Benchmark: Test 4

Test 4 is a flat floodplain filled from a breach, and it tests something 8A does not: how fast an advancing flood front travels and how deep the water is behind it. The grey band is the envelope of the 19 published packages; we sit inside it at all nine points along the cross-section, on both our schemes.

This one we built ourselves from the written specification, because the EA report states that no terrain file is supplied — the floodplain is flat at elevation zero and every other parameter is given numerically. That makes it reproducible without the benchmark data package.

Underneath both benchmarks sit five analytic tests where the correct answer is known exactly: volume conservation in a closed basin, steady runoff matching the Manning normal depth to within half a millimetre, still water staying still over an uneven bed, and agreement with an independent reference implementation to 0.027 mm.

If asked why not all eight EA tests: the remaining cases need terrain and boundary files we do not have, and two of them require coupled 1-D sewer or river models, which this tool does not implement. Test 3 in particular would be worth acquiring — it is the case that discriminates shock-capturing schemes, which is exactly our new capability.

SLIDE 9

Resolution

This slide answers a question the EA benchmark report explicitly says is unresolved: whether resolutions finer than 2 m are worth the cost. The report recommends someone test it at 0.5 m. As far as we know nobody had, and we are well placed to because we have a real dense city surveyed at half a metre.

The method matters for fairness. We coarsened the same terrain step by step rather than rebuilding a model at each resolution, so the survey, the land cover and the corridor design are all held fixed and the computational grid is the only thing that changes.

The result is stark. At 2 m — the resolution the benchmark itself uses — the corridor's measured benefit falls from 9.4 % to 0.6 %. A study run at that resolution would conclude the green corridor

does essentially nothing.

Peak velocity meanwhile rises as the grid coarsens, from 4.3 to 14.8 m/s, which is both the wrong direction and physically implausible. Block-averaging manufactures artificial steep gradients between cells while erasing the kerbs and narrow streets that actually convey the water.

Honest caveat if pressed: coarsening an existing terrain is not identical to building a model natively at 2 m, where roughness would normally be recalibrated to compensate. The direction and size of the effect are nonetheless unambiguous.

SLIDE 10

Chapter: Beirut

Section break.

SLIDE 11

Before and after

The observed 25 November 2025 storm, before and after the corridor, with the difference on the right. Blue means drier after.

Two things to point out on the map. First, the effect is not confined to the ribbon itself — the blue extends into the surrounding street network, because the corridor intercepts runoff coming down from the Achrafieh side before it reaches those streets. Second, the corridor follows a natural drainage axis, which is why it works: it is already where the water wants to go.

If asked about the large square feature near the top left: that is a real deep excavation on a construction site, which the LiDAR surveyed and which we model as a basin. It is genuine terrain, not an artefact.

SLIDE 12

Results

These are the headline numbers, measured on the corridor's walkable surface.

In a frequent storm the corridor removes well over half the flooding from the surfaces people actually walk on. In the observed November event, about half. In a 50-year storm, essentially nothing — and that deserves explanation rather than burial, which is the next slide.

The metric matters. About a fifth of the corridor is deliberately dug out as bioswales, rain gardens and terraces. Those hold water on purpose. Counting a full bioretention pond as 'flooding' penalises the design for working, so the headline figure excludes them. Those cells are 20 % of the corridor but 34 % of its wet area, which is why the two ways of measuring diverge.

The bar chart shows the benefit by distance from the corridor. It reaches roughly two blocks — about 25-50 m — before fading, which is a useful planning number: this is a linear intervention with a local catchment, not a city-wide fix.

SLIDE 13

The severe storm, and a correction

This is the most interesting result and the one most likely to be challenged.

In a 50-year storm the corridor stops reducing the flooded area on its own footprint. But it is still doing hydrological work: it absorbs over two thousand cubic metres more water than the same ground did before, and cuts the volume heading for the port by about a tenth. Its swales and ponds fill and hold. The water is on the corridor by design instead of running down the surrounding streets.

So the corridor has two different jobs — keeping its own surface usable, and protecting what is downstream — and only the second survives an extreme event. That is a more useful message for a planner than a single percentage.

The correction on this slide should be stated plainly rather than left for someone to find. The earlier version of this study credited the corridor with roughly tripling infiltration. With a proper wetting-front infiltration model (Green-Ampt) the multiple is about 1.3. The absolute volume gain holds up — around 900 cubic metres in the observed storm, inside the previously published range — but ordinary Beirut ground turns out to absorb far more than the earlier constant-rate model implied, so the corridor's relative contribution is smaller. Better physics, smaller claim.

SLIDE 14

What would make this a validated model --- the ask

This is the slide to deliver carefully, because it is both the honest caveat and the request.

Make the distinction clearly first: the model is VERIFIED — it solves the equations correctly, reproduces analytic solutions, passes two international benchmarks and agrees with an independently written solver to 3.7 cm. What it is not is VALIDATED, because every one of those comparisons is against another model. We have observed rainfall for 25 November 2025, but no measured flood depths, extents or timings anywhere in the study area.

Then make the ask, positively rather than apologetically. This gap cannot be closed by better modelling, more computing time or another benchmark. It can only be closed by measurement, and the measurement needed is genuinely modest: after the next significant storm, photographs and levelled water marks at a few dozen street locations, ideally within a day or two while the marks are still legible. A handful of inexpensive depth loggers at known ponding points would add timing. None of this requires specialist equipment or a large team.

Be explicit about the value: with that data the study reports a measured error against reality rather than an agreement with other models, and its results can carry planning decisions with real confidence. Without it, everything here remains a well-built and well-tested hypothesis.

Offer to help design the campaign — where to place loggers, which streets matter most — since the model already predicts where flooding concentrates and can be used to site the measurements efficiently. That turns the request into a collaboration rather than a favour.

If asked whether the current numbers are usable meanwhile: yes, with care. The study reports differences — before versus after — and a systematic bias affects both scenarios and largely cancels. Absolute depths carry more uncertainty than the comparisons do.

SLIDE 15

Close

Thank you / questions.

SLIDE REF

Key numbers

Corridor results — flooded-area reduction, best-validated configuration (shock-capturing scheme with Green–Ampt infiltration). “Walkable” excludes the fifth of the corridor dug out as swales and ponds, which hold water by design.

Storm	Walkable	Whole ribbon	Extra absorbed	Outflow cut
frequent (T2)	58.1%	38.6%	+911 m ³	18%
observed 25 Nov 2025	52.0%	29.5%	+919 m ³	17%
severe (T50)	-0.7%	-13.5%	+2124 m ³	11%

Validation — against SynxFlow, an independently written full shallow-water solver, on the identical 0.5 m grid and storm.

Scheme	Extent IoU	Correlation	RMSE	Wet-area bias
Fast inertial (3-term)	0.690	0.938	7.8 cm	+27%
Shock-capturing (HLLC)	0.913	0.985	3.7 cm	+1%

Grid resolution — same terrain, coarsened; only the computational grid varies.

Resolution	Cells	Corridor benefit	p99 street velocity
0.5 m	2 285 142	9.4%	4.3 m/s
1.0 m	571 000	2.4%	8.1 m/s
2.0 m	142 500	0.6%	11.9 m/s
4.0 m	35 500	2.4%	14.8 m/s

If you are asked something not covered here, the safe answer is that the model is verified against analytic solutions and agrees closely with an independent solver, but has never been tested against a measured flood in Beirut, because no such measurement exists. Say that rather than improvise.